

201
BSC

Jharkhand University of Technology, Ranchi
Diploma 2nd Semester Examination, 2025 (NEP)

Subject : Fundamentals of Electrical and Electronic Engineering **Subject Code : BSC-201**

Time Allowed : 3 Hours

Full Marks : 70

Pass Marks: 21

Answer in your own words.

Answer any five questions in which Question No. 1 is compulsory.

The figure in the margin indicate full marks.

All questions carry equal marks.

1. Choose the correct answer:

2×7=14

- (i) What does a yellow safety sign indicate?
(a) Mandatory action (b) Warning
(c) Prohibition (d) Emergency escape
- (ii) Which device is used to measure electrical energy?
(a) Voltmeter (b) Ammeter
(c) Wattmeter (d) Energy meter
- (iii) What is the SI unit of electric current?
(a) Volt (b) Ohm
(c) Ampere (d) Watt
- (iv) Which type of earthing is commonly used in residential buildings?
(a) Pipe earthing (b) Plate earthing
(c) Rod earthing (d) Strip earthing
- (v) Which tool is used to strip insulation from wires?
(a) Pliers (b) Screwdriver
(c) Wire stripper (d) Hammer
- (vi) What is the transformation ratio of a transformer?
(a) V_1/V_2 (b) I_1/I_2
(c) N_1/N_2 (d) All of these
- (vii) Which logic gate gives a high output only when inputs are different?
(a) AND (b) OR
(c) XOR (d) NAND

(2)

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2. (a) Explain the importance of PPE in electrical safety with examples. 7+7
(b) Describe the procedure to administer CPR to an unconscious person.
3. (a) Define voltage, current and resistance. Mention their SI units. 7+7
(b) Explain the behaviour of current and voltage in a parallel circuit with examples.
4. (a) Describe the working principle of a single-phase transformer. 7+7
(b) Compare the features and applications of DOL and Star-Delta starters.
5. (a) Explain the function of RCCB and ELCB in electrical protection.
(b) Draw and explain the wiring diagram for controlling one lamp from two different places. 7+7
6. (a) Describe the working of a bridge rectifier with and without capacitor filter. 7+7
(b) Explain the operation of a relay as an actuator in a control circuit. 3-5x4
7. Write short notes on *any four* of the following:
(a) Types of fire extinguishers and their applications
(b) CRO
(c) Types of batteries and their selection criteria
(d) Rectifier
(e) Microcontroller applications in automation
(f) Ripple factor